

Jeffery Liu

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Penultimate-year Mechatronics Engineering student specialising in mechanical design, testing and production. At Holmes Solutions I took an apparatus upgrade from research through CAD, FEA and drawings to a built design that passed full-scale testing, and co-authored the resulting conference paper. Backed by real production skill in diagnostics and repair, machining and fabrication, plus over three years running my own licensed vehicle dealership and growing electronics and controls experience from UC Motorsport (FSAE). Seeking a 2026/27 summer internship in mechanical or mechatronics engineering.

EDUCATION

University of Canterbury, BE(Hons) Mechatronics Engineering 2024 – present

- Third year, expected graduation 2027. Offered two competitive UC entrance international scholarships.

EXPERIENCE

Mechatronics Engineering Intern, Holmes Solutions Nov 2025 – Feb 2026

Roadside & Marine sector, R&D consultancy (453.5 hours)

- Investigated premature mooring rope failures and designed a **sleeve retrofit** rated above **50 tonnes**.
- Verified the design with hand calculations and **SolidWorks FEA**.
- Produced **construction drawings** and a **staged build plan**, then **sourced contractors** and material suppliers.
- Design was built and passed full-scale testing; co-authored the resulting **conference paper**.
- Assisted **dynamic testing** setup; practised **welding**, **machining** and **plasma cutting**.

Reference: Ray Keane, Senior Engineer, ray.keane@holmessolutions.com

Founder & Licensed Vehicle Dealer, Jeff's Auto Retail 2023 – 2026

- Registered as a sole trader at 18 as a **self-funded** international student.
- Ran the business from my own workshop instead of a yard, prioritising **fast turnover**.
- Sourced, diagnosed, repaired and resold **35+ cars** solo, to WOF standard.
- Handled all **business operations**: pricing, listings, customer relations, bookkeeping, tax and compliance.

Junior Vehicle Dynamics Engineer, UC Motorsport (FSAE) Apr 2026 – present

Junior lead on the tyre-temperature sensing system project

- Researched requirements and selected **IR pyrometry** for onboard tyre-temperature logging.
- Owned schematic, **PCB layout** (KiCad) and component selection for an **ESP32-based** sensor module.
- Self-taught **SMT**, assembled the board via pick-and-place and reflow, and wrote **CircuitPython** firmware.
- Designed a 3D-printed clip-on housing built to handle 2.5+ G; first track test produced clean, filtered data.

Performance Consultant & Tuning Support, Custom Culture & Tuning 2025 – present

- Consult clients on parts and performance; support an Auckland tuner with **datalogging** and adjustments.

Reference: Amit Lal, Tuner/Business Owner, cct.tune@gmail.com

Bicycle Technician & Maintenance Tutor, Adventure Cycles 2019 – 2023

- Diagnosed, serviced and repaired all bicycle components, and taught public **maintenance workshops**.

Bike Mechanic, Evo Cycles / Pitcrew Cycles Oct 2021 – Jan 2023

- Assembled bikes and carried out mechanical servicing during school holidays.

Workshop Apprentice, Steelguard Ltd Mar 2023 – Aug 2023

- Assisted during a post-flood relocation, gaining machine shop and welding experience.

PROJECTS

Turbocharged Engine Rebuild & Swap, 2006 Subaru Legacy GT

2025 – 2026

- Diagnosed an internal engine failure and rebuilt with stronger internals to hit a target power goal.
- Validated through datalogging and tuner collaboration; now daily driven.

Line Follower Robot, Team University Project

2025

- Led mechanical design, drivetrain and silicone tyres production, custom PCB with PID control.
- Set a practice lap fast enough to win; finished 3rd after a late motor failure.

Pick and Place CNC Machine, Team University Project

2025

- Team of five: CAD-modelled and built a sub-\$200 pick-and-place CNC with a waterjet-cut aluminium frame.
- Delivered for \$128.85, assembled in 1 hour, positioning within 0.3 mm.

Fixed Wing UAV

2025

- Designed a reflexed airfoil with root-to-tip washout to improve stall behaviour on a flying-wing layout.
- Selected the propulsion and electronics; built a hot-wire cutter to cut the foam wing cores to profile.

RoboCup Autonomous Robot, Team University Project

2026 – present

- Team of three building an autonomous robot for a timed arena challenge; proposed the design concept.
- Leading mechanical design; drum intake, ToF obstacle response and sorting mechanism in development.

PLC Elevator Control System, Team University Project

2025

- Programmed an Omron PLC in structured text & ladder logic; request queue, state machine and PI control.

SKILLS

Mechanical & Production

- SolidWorks (FEA)
- COMSOL
- GD&T
- DFMA
- CAM (3D printing, CNC)
- MIG & arc welding
- Plasma & oxy cutting
- Machine tool operation
- Diagnostics & repair
- Structural verification

Electronics & Hardware

- PCB design (KiCad)
- Soldering
- SMT
- Hardware testing
- Electrical diagnosis
- Auto & marine wiring

Controls & Embedded

- PLC (ladder logic)
- Embedded systems (C, Python)
- Control theory (PID)
- Datalogging
- MATLAB/Simulink
- Git / version control

Business & Communication

- Small business management
- Bilingual (English/Mandarin)
- Technical reporting
- Contractor coordination
- Customer consultation

Developed through university coursework, an R&D engineering internship, UC Motorsport, and hands-on vehicle and business experience.

ADDITIONAL INFORMATION

Work Rights: New Zealand Permanent Resident, full right to work in New Zealand with no restrictions.

Driver's Licence: Full New Zealand licence.

Qualifications: NZ Workplace First Aid Certificate, Unit standards 6400, 6401, 6402, March 2026.

Hobbies: cars, boating, fishing, camping, snowboarding, cooking.